The Keystroke-Level Model for User Performance Time with Interactive Systems

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ABSTRACT

It is not common practice today for system designers to deal systematically with the issues of user-computer performance. One reason is the lack of appropriate analysis tools. An easy-to-use model—the Keystroke Model—is proposed for predicting the time it will take expert users to execute given tasks on a system. The Model is based on counting keystrokes and other low-level operations, including the user's mental preparations and the system's responses. Methods for executing tasks are coded in terms of these operations, and standard times for the operations are then summed up to give predictions. Heuristic rules are given for predicting where mental preparations occur in the methods. Keystroke Model predictions were tested against data from 28 users, on 10 systems, and over 14 task types. The root-mean-square prediction error was 21% for individual tasks—and much better for collections of tasks. An example illustrates how the Keystroke Model can be used to give parametric predictions and how sensitivity analysis can be used to redeem conclusions in the face of uncertain assumptions. Finally, the Keystroke Model is compared to several simpler versions, which trade ease-of-use for accuracy.

Key words and phrases:

human-computer interface, human-computer interaction, user model, user performance, cognitive ergonomics, human factors, systems design

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The design and evaluation of interactive computer systems should take into account the total performance of the combined user-computer system. Such an account would reflect the psychological characteristics of users and their interaction with the task and the system. This rarely occurs in any systematic and explicit way. The causes of this failure may lie partly in attitudes towards the possibility of dealing successfully with psychological factors. The causes may lie partly in the belief that intuition, subjective experience, and anecdote form the only possible bases for dealing with total performance. Whatever may be true of these more global issues, one major cause is the absence of good analysis tools for assessing combined user-computer performance.

This paper presents one such tool: a simple model for the time it takes a user to perform a task with a given method on an interactive computer system. This model is simple enough to be applied in practical design and evaluation situations, yet appears accurate enough to be of genuine help.

The model addresses only a single, specific aspect of performance. To put this aspect into perspective, note first that there are many important dimensions to the performance of a user-computer system:

—*Time.* How long does it take a user to accomplish a given set of tasks using the system?
—*Errors.* How many errors does a user make and how serious are they?
—*Learning.* How long does it take a novice user to learn how to use the system to do a given set of tasks?
—*Functionality.* What range of tasks can a user do in practice with the system?
—*Recall.* How easy is it for a user to recall how to use the system on a task that he has not done for some time?
—*Concentration.* How many things does a user have to keep in mind while using the system?
—*Fatigue.* How tired do users get when they use the system for extended periods?
—*Acceptability.* How do users subjectively evaluate the system?

Next, note that there is *no single kind of user.* Users vary along many dimensions:

—Their *extent of knowledge* of the different tasks.

—Their *knowledge of other systems* (which may have positive or negative effects on the performance in the system of interest).

—Their *motor skills* on various input devices (e.g., typing speed).

—Their general *technical ability* in using systems (e.g., programmers vs. non-programmers).

—Their *experience* with the system, i.e., whether they are: *novice users*, who know little about the system; *casual users*, who know a moderate amount about the system and use it at irregular intervals; or *expert users*, who know the system intimately and use it frequently.

Finally, note there is *no single kind of task.* This holds especially in interactive systems, which are expressly built around a command language to permit a wide diversity of tasks to be accomplished. The qualitatively different tasks performable by a modern text editor, for instance, runs to the hundreds.

All aspects of performance, all types of users, and all kinds of tasks are important. However, no uniform approach to modeling the entire system in a simple way appears possible at this time. Thus, of necessity, the model to be presented is specific to one aspect of the total user-computer system: *how long it takes expert users to perform routine tasks.*

The model we present here is extremely simple, yet it seems amazingly effective. The central idea is: *The time for an expert to do a task on an interactive system is simply the time it takes to do the keystrokes.* Therefore, just write down the method for the task, count the number of keystrokes required, and multiply by the time per keystroke to get the total time. This idea is a little too simplistic,† as we will show; but the model we present preserves most of its appealing simplicity. Hence, we dub it the "Keystroke Model".

Section 1 of this paper formulates the time prediction problem more precisely. Section 2 lays out the Keystroke Model. Section 3 provides some empirical validation for the model. Section 4 illustrates how the model can be applied in practice. And Section 5 analyzes some simpler versions of the model.

†Perhaps for fear of appearing too simplistic, no one has taken this basic idea seriously enough to test it. In fact, only as we were finishing up this paper did we discover another paper (Embly et al., 1978) that formally proposed a model based on keystroke counting, though it did no testing of the model.

1. The Time Prediction Problem

The prediction problem that we will address is as follows:

Given —A task (possibly involving several subtasks).
—The command language of a system.
—The motor skill parameters of the user.
—The response time parameters of the system.
—The method used for the task.

Predict —The time an expert user will take to execute the task using the system, providing he uses the method without error.

Several aspects of this formulation need explication, especially the stipulations about execution, methods, and the absence of error.

1A. Unit Tasks and Execution Time

Given a large task, such as editing a large document, a user will break it into a series of small, cognitively manageable, quasi-independent tasks, which we call *unit tasks* (Card, Moran, & Newell, in press). The task and the interactive system influences the structure of these unit tasks, but they appear to owe their existence primarily to the memory limits on human cognition. The importance of unit tasks for our analysis is that they permit the time to do a large task to be decomposed into the sum of the times to do its constituent unit tasks.†

For our purposes here, a unit task has two parts: (1) *acquire* the task and (2) *execute* the task acquired. During acquisition the user builds a mental representation of the task, and during execution the user calls on the system facilities to accomplish the task. The total time to do a unit task is the sum of the time for these two parts:

$$T_{task} = T_{acquire} + T_{execute}$$

The acquisition time for a unit task depends on the characteristics of the larger task situation in which it occurs. In a manuscript interpretation situation, in which unit tasks are read from a marked-up page or from written instructions, it takes about 2–3 sec to acquire each unit task. In a routine design situation, in which unit tasks are generated in the user's mind, it takes about 5–30 sec to acquire each unit task. In a creative composition situation, it can take even longer.

†Not all tasks have a unit-task substructure. For example, inputting an entire manuscript by typing permits a continuous throughput organization. We do not intend to present an exhaustive task analysis, only to make clear this particular model.

The execution of a unit task involves calling the appropriate system commands. This rarely takes over 20 sec (assuming the system has a reasonably efficient command syntax). If a task requires a longer execution time, the user will likely break it into smaller unit tasks.

We have formulated the prediction problem to predict only the execution time of unit tasks, not the acquisition time. This is the part of the task that the system designer has most direct control over (i.e., by manipulating the system's command language), so its prediction suffices for many practical purposes. Task acquisition times are highly variable, except in special situations (such as the manuscript interpretation situation); and we can say little yet about predicting them.

Two important assumptions underlie our treatment of execution time. First, execution time is the same no matter how a task is acquired. Second, acquisition time and execution time are independent, i.e., squeezing execution time (by making the command language more efficient) does not affect acquisition time. These assumptions are no doubt false at a fine level of detail, but the error they produce is probably well below the threshold needed for practical work.

1B. Methods

A *method* is a sequence of system commands for executing a unit task that forms a well-integrated ("compiled") segment of a user's behavior. It is characteristic of an expert user that he has one or more methods for each type of unit task that he encounters, and he can quickly (in about a second) choose the appropriate method in any instance. This is what makes expert user behavior routine, as opposed to novice user behavior, which is distinctly non-routine.

Methods can be specified at several levels. A user actually knows a method at all its levels, from a general system-independent functional specification, down through the commands in the language of the computer system, to the keystrokes and device manipulations that actually communicate the method to the system. Models can deal with methods defined at any of these levels (Card, Moran, & Newell, 1976). The Keystroke Model adopts one specific level—the "keystroke" level—to formalize the notion of a method, leaving all the other levels to be treated informally.

Many methods can exist that achieve a given task. In general such methods bear no systematic relationship to each other (except that of attaining the same end). Each can take a different time to execute and the differences can be large. Thus, in general, if the method is unknown, reasonable predictions of execution time are not possible. For this reason, the proper prediction problem is the one posed at the beginning of the section: predict the time given the method.

1C. Error-Free Execution

The Keystroke Model assumes that the user makes no errors during execution. Expert users do, of course, make errors. Up to a fourth of an expert's time can be spent correcting errors, though users vary in their trade-off between speed and errors. We are simply ignoring the error tasks and only predicting the error-free tasks, for we do not know how to predict where and how often errors occur. On the other hand, given the method by which an error is handled, the model will produce a prediction of how long it will take. Indeed, experts handle most (but not all) errors in routine ways, that is, according to fixed, available methods.

2. The Keystroke Model

We lay out the primitive operators for the Keystroke Model and give a set of heuristics for coding methods in terms of these operators. Then we present a few examples of method encoding.

2A. Operators

The Keystroke Model asserts that execution time consists of four different physical-motor operators (**K**, **P**, **H**, and **D**) and one mental operator (**M**) by the user, plus a response operator (**R**) by the system. These operators are listed in Figure 1. Execution time is simply the sum of the time for each of the operators:

$$T_{execute} = T_K + T_P + T_H + T_D + T_M + T_R \,. \qquad (1)$$

Most operators are assumed to take a constant time for each occurrence, i.e., $T_\alpha = n_\alpha t_\alpha$ (operators **D** and **R** are treated somewhat differently, see below).

The most frequently used operator is **K**, which consists of a keystroke or a button push (on a typewriter keyboard or any other button device). **K** refers to keys, not characters (e.g., hitting the SHIFT key counts as a separate **K**). The average time for **K**, t_K, will be taken to be the standard typing rate, as determined by standard one-minute typing tests. This is an approximation in two respects. First, keying time is different for different keys and key devices. Second, the time for immediately-caught typing errors (involving BACKSPACE and rekeying) should be folded into t_K. Thus, the preferred way to calculate t_K from a typing test is to divide the total time taken in the test by the total number of non-error keystrokes, which gives the *effective* keying time. We accept both these approximations in the interest of simplicity.

People can differ in their typing rate by as much as a factor of 15. The range of typing speeds is given in Figure 1. Given a population of users, an appropriate t_K can be selected from this range. If a user population has users with large t_K differences; then the population should be partitioned and analyzed separately, since the different classes of users will likely use different methods.

The operator **P** is for pointing to a target on a display with a "mouse", a device that is rolled around a table on its wheels to guide the display's cursor. Pointing time for the mouse varies as a function of the distance to the target (D) and the size of the target (S) according to Fitts's Law (Card, English, & Burr, 1978):

$$t_P = .8 + .1 \log_2(D/S + .5) \text{ sec.}$$

Operator	Description and Remarks	Time (sec)
K	A keystroke or button press. Pressing the SHIFT or CONTROL key counts as a separate K operation. Time varies with the typing skill of the user; the following shows the range of typical values:	
	Best typist (135 wpm)	.08[a]
	Good typist (90 wpm)	.12[a]
	Average skilled typist (55 wpm)	.20[a]
	Average non-secretary typist (40 wpm)	.28[b]
	Typing random letters	.50[a]
	Typing complex codes	.75[a]
	Worst typist (unfamiliar with keyboard)	1.20[a]
P	Pointing to a target on a display with a mouse. The time to point varies with distance and target size according to Fitts's Law. The time ranges from .8 to 1.5 sec, with 1.1 being an average time. This operator does *not* include the button press that often follows (.2 sec).	1.10[c]
H	Homing the hand(s) on the keyboard or other device.	.40[d]
$D(n_D, l_D)$	Drawing n_D straight-line segments having a total length of l_D cm. This is a very restricted operator; it assumes that drawing is done with the mouse on a system that constrains all lines to fall on a square .56 cm grid. Users vary in their drawing skill; the time given is an average value.	$.9n_D + .16l_D$[e]
M	Mentally preparing for executing physical actions.	1.35[f]
$R(t)$	Response *by the system.* This takes different times for different commands in the system. These times must be input to the model. The response time counts only if it causes the user to wait.	t

[a] See Devoe (1967).
[b] This is the average typing rate of the non-secretary subjects in the experiment described in Section 3A.
[c] See Card, English, & Burr (1978).
[d] See Card, Moran, & Newell (1976); Card, English, & Burr (1978).
[e] The drawing time function and the coefficients were derived from least squares fits on the drawing test data from the four MARKUP subjects. See Sections 2A and 3A.
[f] The time for M was estimated from the data from experiment described in Section 3A. See Section 3B1.

Figure 1. The Operators of the Keystroke Model

Begin with a method encoding that includes all physical operations and response operations. Use Rule 0 to place candidate **M**s, and then cycle through Rules 1 to 4 for each **M** to see whether it should be deleted.

Rule 0. Insert **M**s in front of all **K**s that are not part of argument strings proper (e.g., text or numbers). Place **M**s in front of all **P**s that select commands (not arguments).

Rule 1. If an operator following an **M** is *fully anticipated* in an operator just previous to **M**, then delete the **M** (e.g., **PMK** → **PK**).

Rule 2. If a string of **MK**s *belong to a cognitive unit* (e.g., the name of a command), then delete all **M**s but the first.

Rule 3. If a **K** is a *redundant terminator* (e.g., the terminator of a command immediately following the terminator of its argument), then delete the **M** in front of it.

Rule 4. If a **K** *terminates a constant string* (e.g., a command name), then delete the **M** in front of it; but if the **K** terminates a variable string (e.g., an argument string), then keep the **M** in front of it.

Figure 2. Heuristic Rules for Placing the M Operations

The fastest time according to this equation is .8 sec, and the longest likely time ($D/S=128$) is 1.5 sec. Again, to keep the model simple we will use a constant time of 1.1 sec for t_P. Often pointing with the mouse is followed by pressing one of the buttons on the mouse. This key press is not part of **P**; it is represented by a **K** following the **P**. The mouse is an optimal pointing device as far as time is concerned, but the t_P for other analog pointing devices (such as a joysticks and lightpens) is about the same (Card, English, & Burr, 1978).

When there are different physical devices for the user to operate, he will have to move his hands between them as needed. This hand movement, including the fine positioning adjustment of the hand on the device, is represented by the **H** ("homing") operator. From previous studies (Card, Moran, & Newell, 1976; Card, English, & Burr, 1978), we assume a constant t_H of .4 sec for movement between any two devices.

The **D** operator represents manually drawing a set of straight-line segments using the mouse. **D** takes two parameters, the number of segments (n_D) and the total length of all segments (l_D). $t_D(n_D,l_D)$ is a linear function of these two parameters. The coefficients of this function are different for different users, and Figure 1 gives an average value for them. Note that this is a very specialized operator. Not only is it restricted to the mouse, but also it assumes that the drawing system constrains the cursor to lie on a .56 cm grid. This allows the user to draw straight lines fairly easily, but we would expect t_D to be different for different grid sizes. We make no claim for the generality of these times or for the form of the drawing time function. However, inclusion of one instance of a drawing operator serves to indicate the wide scope of the model.

The user spends some time "mentally preparing" to execute many of the physical operators (just described), e.g., he decides which command to call or whether to terminate an argument string. These mental preparations are represented by the **M** operator, which we estimate takes 1.35 sec on the average (see Section 3B1). The use of a single mental operator is, again, a deliberate simplification.

Finally, the Keystroke Model represents the system response time by the **R** operator. Response times are different for different systems, for different commands within a system, and for different contexts of a given command. The Keystroke Model does not embody a theory of system response time. The response times must be input to the Keystroke Model in the form of parameters to the **R** operations, which are placeholders for these input times.

The **R** times are counted only when they require the user to *wait for the system*. When a system response is followed by a **K** and the system does not allow type-ahead, the user must wait until the response is complete; and the response time counts as an **R**. However, when an **M** operation follows a response, it is not counted unless it is over 1.35 sec, since the expert user can completely overlap the **M** operation with the response time. Response times can also overlap with task acquisition. When a response is counted as an **R**, only the non-overlapping portion of the response time is given as a parameter to an **R**.

2B. Encoding Methods

Methods are represented as sequences of Keystroke Model operations. We will introduce the notation with examples. Suppose that there is a command named PUT in some system and that the method for executing it is to type its name followed by the RETURN key. This method is coded by simply listing the operations in sequence: **MK[P] K[U] K[T] K[RETURN]**, which we abbreviate as **M 4K[P U T RETURN]**. In this notation we allow descriptive notes (such as key names) in square brackets. If, on the other hand, the method to call the PUT command were to point to its name in a menu and press the RED mouse button, we would have: **H[mouse] MP[PUT] K[RED] H[keyboard]**.

As an example, consider the text editing task (called T1) of replacing a 5-letter word with another 5-letter word, where this replacement takes place one line below the previous modification. The method for executing task T1 in a line-oriented editor called POET (see Section 3) can be described as follows:

Method for Task T1-POET:	
Jump to next line	MK[LINEFEED]
Call Substitute command	MK[S]
Specify new 5-digit word	5K[word]
Terminate argument	MK[RETURN]
Specify old 5-digit word	5K[word]
Terminate argument	MK[RETURN]
Terminate command	K[RETURN]

Using the operator times from Figure 1 and assuming the user is an average skilled typist (i.e., t_K = .2 sec), we can predict the time it will take to execute this method:

$$T_{execute} = 4t_M + 15t_K = 8.4 \text{ sec}$$

This method can be compared to the method for executing task T1 on another editor, a display-based system called DISPED (see Section 3):

Method for Task T1-DISPED:	
Reach for mouse	H[mouse]
Point to word	P[word]
Select word	K[YELLOW]
Home on keyboard	H[keyboard]
Call Replace command	MK[R]
Type new 5-digit word	5K[word]
Terminate type-in	MK[ESC]

$$T_{execute} = 2t_M + 8t_K + 2t_H + t_P = 6.2 \text{ sec}$$

Thus, we predict that the task will take about two seconds longer on POET than on DISPED. In Section 3 we will see how accurate such predictions are.

The methods above are quite simple in that they are unconditional sequences. More complex or more general tasks are likely to have multiple methods and/or conditionalities within methods for accomplishing different versions of the task. For example, in a DISPED-like system the user often has to scroll the text before being able to point to the desired target. We can represent this as follows:

.4(MP[SCROLL-BAR] K[RED] R(.5)) P[word] K[YELLOW] .

Here we assume a specific situation where the average number of scroll jumps per selection is .4 and that the average system response time for a scroll jump is .5 sec. From this we can predict the average selection time:

$$T_{execute} = .4t_M + 1.4t_K + 1.4t_P + .4(.5) = 2.6 \text{ sec}$$

For more complex contingencies, we can put the operations on a flow graph and label the paths with their frequencies.

When there are alternative methods for doing a *specific* task in a given system, we have found that expert users will in general use the most efficient method, i.e., the method taking the least time. Thus, in making predictions we can use the model to compute the times for the alternative methods and predict that the fastest method will be used (if the alternatives take about the same time, it doesn't matter which method we predict). This optimality assumption holds, of course, only if the users are familiar with the alternatives, which is usually true of experts, unless some alternatives are esoteric. The assumption is helped by the tendency of optimal methods to be the simplest.

2C. Heuristics for the M Operator

M operations represent acts of mental preparation for applying subsequent physical operations. Their occurrence does not follow directly from the method as defined by the command language of the system, but from the specific knowledge and skill of the user. The Keystroke Model provides a set of rules for placing **M**'s in the method encodings. These rules embody psychological assumptions about the user and are necessarily heuristic, especially given the simplicity of the model. They should be viewed simply as guidelines. The basic principle is: *Place an anywhere a decision must be made, except where the decision can be overlapped with other operations that do not depend on it.* The rules given in Figure 2 define a procedure that implements this principle.

The procedure begins with an encoding that contains only the physical operations (**K**, **P**, **H**, and **D**). First, all candidate **M**'s are inserted into the encoding according to Rule 0, which is a heuristic for identifying all possible decision points in the method. Rules 1 to 4 are then applied to each candidate **M** to see if it should be deleted.

There is a single principle behind all the deletion heuristics. Methods are composed of highly integrated submethods ("subroutines") that show up over and over again in different methods. We will call them *method chunks* or just *chunks*, a term common in cognitive psychology (Simon, 1974). The user cognitively organizes his methods according to chunks, which usually reflect syntactic constituents of the system's command language. Hence, the user prepares mentally for the next chunk, not just the next operation. It follows that in executing methods the user is more likely to pause between chunks than within chunks. The rules attempt to identify method chunks.

Rule 1 asserts that when an operation is fully anticipated in another operation, they belong in a chunk. A common example is pointing with the mouse and then pressing the mouse button to indicate a selection. The button press is fully anticipated during the pointing operation, and there is no pause between them (i.e., **PMK** becomes **PK** according to Rule 1). This anticipation holds even if the selection indication is done on another device (e.g., the keyboard or a foot pedal). Rule 2 asserts that obvious syntactic units, such as names, are chunks when they have to be typed out.

The last two heuristics deal with syntactic terminators. Rule 3 asserts that the user will bundle up redundant terminators into a single chunk. For example, in the POET example in Section 2B a RETURN is required to terminate the second argument and then another RETURN to terminate the command; but any user will quickly learn to simply hit a double RETURN after the second argument (i.e., **MKMK** becomes **MKK** according to Rule 3). Rule 4 asserts that a terminator of a constant-string chunk will be assimilated to that chunk. The most common example of this is in systems that require a terminator, such as RETURN, after command names; the user learns to immediately follow the command name with RETURN.

It is clear that these heuristics do not capture the notion of method chunks precisely, but are only crude approximations. Further, their application is ambiguous in many situations, e.g., whether something is "fully anticipated" or is a "cognitive unit". What can we do about this ambiguity? Better general heuristics will help in reducing this ambiguity. However, some of the variability in what are chunks stems from a corresponding variability in expertness. Individuals differ widely in their behavior; their categorization into *novice*, *casual*, and *expert* users provides only a crude separation and leaves wide variation within each category. One way in which experts will differ is in what chunks they have (see Chase & Simon, 1973, for related evidence). Thus, some of the difficulties in placing **M**'s is unavoidable because not enough is known (or can be known in practical work) about the experts involved.

Expertise is of many sorts. Besides the general expertise that comes from much experience with a system over a wide range of tasks, there is also local expertise that comes from repeating the same unit task over and over again in immediate succession. Users will get faster on such tasks with repetition, though for practical purposes they will reach a plateau. Part of the increase in speed can be represented in the Keystroke Model as successively squeezing out **M** operations.

3. Empirical Validation of the Keystroke Model

To determine how well the Keystroke Model actually predicts performance times we ran an experiment in which calculations from the model were compared against measured times for a number of different tasks, systems, and users.

3A. Description of the Experiment

A total of 1280 user-system-task interactions were observed, comprised of various combinations of 28 users, 10 systems, and 14 tasks.

Systems. The systems were all typical application programs available locally (at Xerox PARC) and widely used by both technical and non-technical users. Some of the systems are also widely used nationally. Three of the systems were text editors, three were graphics editors, and five were executive subsystems. The systems are briefly described in Figure 3.

Together these systems display a considerable diversity of user interface techniques. For example, POET, one of the text editors, is a typical line-oriented system, which uses first-letter mnemonics to specify commands and search strings to locate lines. In contrast, DRAW, one of the graphics systems, displays a menu of graphic icons on the CRT display to represent the commands, which the user selects by pointing at with the mouse.

Tasks. The 14 tasks performed by the users (see Figure 4) were also diverse, but typical. Users of the editing systems were given tasks ranging from a simple word substitution to the more difficult task of moving a sentence from the middle to the end of a paragraph. Users of the graphics systems were given tasks such as adding a box to a diagram or deleting a box (but keeping a line which overlapped it). Users of the executive subsystems were given tasks such as transferring a file between computers or examining part of a file directory.

Task-system methods. In all there were 32 task-system combinations: 4x3=12 for the text editors, 5x3=15 for the graphics systems, and one task each for the five executive subsystems. For each task-system combination, the most efficient "natural" method was determined (by consulting experts) and then coded in Keystroke Model operations. For example, the methods for T1-POET and T1-DISPED are given in Section 2B.

Experimental design. The basic design of the experiment was to have ten versions of each task on each system done by four different users, giving 40 observed instances per task-system. No user was observed on more than one system to avoid transfer effects. Four tasks were observed for each of the text editing systems, and five tasks for each of the graphics systems, and one task for each of the executive subsystems.

System	Description
Text Editors	
POET[a]	Line-oriented with relative line numbers.
SOS[b]	Line-oriented with "sticky" line-numbers.
DISPED[e]	Display-oriented; full-page; uses mouse for pointing.
Graphics Systems	
MARKUP[c]	Uses mouse to draw and erase lines on a bitmap display; commands selected from a hidden menu, which must be re-displayed each time.
DRAW[e]	Lines defined by pointing with mouse to end points; commands selected with mouse from a menu.
SIL[e]	Lines defined by pointing with mouse to end points; boxes defined by pointing to opposite vertices; commands selected by combinations of mouse buttons.
Executive Subsystems	
LOGIN[d]	TENEX command for logging in.
FTP[e]	Program for transferring files between computers.
CHAT[e]	Program for establishing a "teletype" connection between two computers.
DIR[d]	TENEX command for printing a file directory; has a subcommand mode.
DELVER[d]	TENEX command for deleting old versions of a file.

[a]POET is a dialect of the QED editor (Deutsch & Lampson, 1967).
[b]See Savitsky (1969).
[c]See Newman & Sproull (1979), Chapter 17.
[d]See Myer & Barnaby (1973).
[e]Experimental systems local to Xerox PARC, designed and implemented by many individuals, including: Roger Bates, Patrick Baudelaire, David Boggs, Butler Lampson, Charles Simonyi, Robert Sproull, Edward Taft, and Chuck Thacker.

Figure 3. Systems Measured in the Experiment

Editing Tasks (used for POET, SOS, DISPED)

T1. Replace one 5-letter word with another (one line from previous task)

T2. Add a 5th character to a 4-letter word (one line from previous task)

T3. Delete a line, all on one line (eight lines from previous task)

T4. Move a 50-character sentence, spread over two lines, to the end of its paragraph (eight lines from previous task)

Graphics Tasks (used for MARKUP, DRAW, SIL)

T5. Add a box to a diagram

T6. Add a 5-character label to a box

T7. Reconnect a 2-stroke line to a different box

T8. Delete a box, but keep an overlapped line

T9. Copy a box

Executive Tasks

T10. Phone computer and log in (4 char name, 6 char password)

T11. Transfer a file to another computer, renaming it

T12. Connect to another computer

T13. Display a subset of the file directory with file lengths

T14. Delete old versions of a file

Figure 4. Tasks for the Experiment

Subjects. There were in all 28 different users (some technical, some secretarial): 12 for the editing systems, 12 for the graphics systems, and 4 for the executive subsystems. All were experts in that they had used the systems for months in their regular work and had used them recently, but no attempt was made to assess their expertness systematically.

Experimental procedure. Each user was first given five one-minute typing tests to determine his keystroke time, t_K. In addition, users of MARKUP (the only system to require manual drawing) were given a series of drawing tasks to determine the parameters of their drawing rate (as discussed in Section 2A).

After the preliminary tests, the user was given a small number of practice problems of the sort to be tested and were told the method to use (see above). In most cases, the methods presented were what users claimed they would have used anyway; in other cases, the method was easily adopted. Users practiced tasks until they were judged to be at ease and using the correct method; this was usually accomplished in three or four practice trials on each task type.

After practicing, the user proceeded to the main part of the experiment. The user was given a notebook containing several manuscript pages with the tasks to be done marked in red ink. Text editing and graphics tasks appeared in randomized order. Executive subsystem tasks were always in the order T11, T12, T13, T14. All ten instances of task T10 were done in succession.

Each experimental session, lasting approximately 40 minutes, was videotaped and the user's keystrokes recorded automatically. Time stamps on the videotaped record and on each keystroke allowed a protocol to be constructed in which the time of each event was known to about .033 sec. This protocol is the basic data from which the results below are derived.

3B. Results of the Experiment

Each task instance in the protocols was divided into acquisition time and execution time (cf. Section 1A) according to the following definitions. Acquistion time began when the user first looked over to the manuscript to get instructions for the next task and ended when the user started to perform the first operator of the method. Execution time began at that point and ended when the user looked over to the notebook for the next task. On the protocol the first measured time at the beginning of an execution is always the end of the first K of the method. Thus, operationally, the beginning of execution time was estimated by subtracting from this first K time the operator times (taken from Figure 1) for this first K plus all the operators that occurred before it.

Those tasks on which there were significant errors (i.e., other than typing errors) or in which the user did not use the prescribed method were excluded from further consideration. After this exclusion, 855 (69%) of the task instances remained as observations to be matched against the predictions. No analysis was made of the excluded tasks.

The resulting observed times for task acquisition and execution were stable over repetition. There was no statistical evidence for times decreasing (learning) or increasing (fatigue) with repetition.

3B1. Calculation of Execution Time

Execution time was calculated using the method analysis for each task-system combination together with estimates of the times required for each operator. All times, except for the mental preparation time, were taken from sources outside of the experiment. Pointing time, t_P, and homing time, t_H, were taken from Figure 1. Typing time, t_K, and drawing time, $t_D(n_D, l_D)$, were estimated from the typing and drawing tests by averaging the times of the four users involved in each task-system. System response time, T_R, for each task-system was estimated from independent measurements of the response times for the various commands required in each method. For task T10, logging in to a computer, pushing a button on the telephone was assumed to take time t_K. Moving the telephone receiver to the computer terminal modem was estimated to take .7 sec, using the MTM system of times for industrial operations (Maynard, 1971).†

Mental preparation time, t_M, was estimated from the experimental data itself. First, the total mental time for each method was estimated by removing the predicted time for all physical operations from the observed execution time. Then t_M was then estimated by a least squares fit of the estimated mental times as a function of the predicted number of **M** operations. The result was t_M = 1.35 sec (R^2 = .84, standard error of estimate = .11 sec, standard error about the regression line = 2.48 sec). A rough estimate of the SD of t_M is 1.1 sec, which indicates that the **M** operator has the characteristic variability of mental operators (Card, Moran, & Newell, in press).

Execution times for each task-system combination were calculated by formula (1) in Section 2. The calculations of the execution times are summarized in Figure 5, which also gives the observed execution times from the experiment for comparison.

3B2. Execution Time

The predicted execution times are quite accurate. This can be seen in Figure 6, which plots the predicted versus observed data from Figure 5. The scales are logarithmic, since prediction error appears to be roughly proportional to duration. The root mean square (RMS) error is 21% of the average predicted execution time. This accuracy is about the best that can be expected from the Keystroke Model, since the methods used by the subjects were controlled by the experimental procedure. The 21% RMS error is comparable to the 20-30% we have obtained in other studies on text editing with more elaborate models that also predict the method (Card, Moran, & Newell, 1976).

The distribution of relative prediction errors is fairly evenly spread, as an analysis of Figure 6 will show. No particular systems or tasks make excessively large contributions. Predictions are not consistently positive or negative for systems or tasks, except that all

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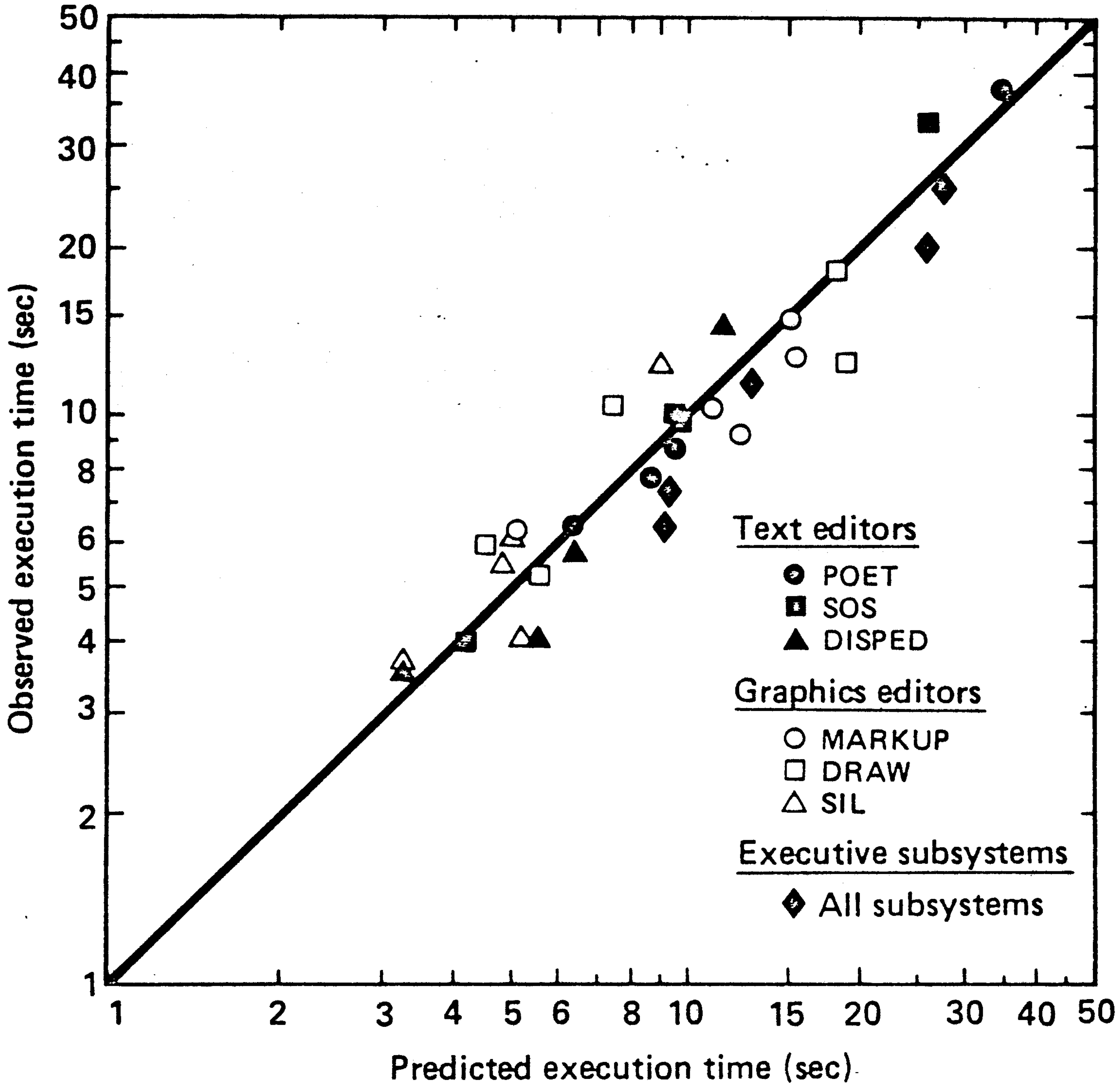
†One point of task T10 is to illustrate that the Keystroke Model can be extended by using existing catalogs of physical operators.

Task-System	Calculated[a]									Observed	Pred. Error
	t_K[b] (sec)	n_M	n_K	n_H	n_P	n_D	l_D (cm)	T_R (sec)	$T_{execute}$ (sec)	$T_{execute}$ $M \pm SE$ (N)[c] (sec) (sec)	
T1-POET	.23	4	15	--	--	--	--	--	8.8	7.8 ± 0.9(27)	11%
T1-SOS	.22	4	19	--	--	--	--	--	9.6	9.6 ± 0.8(31)	1%
T1-DISPED	.23	2	8	2	1	--	--	--	6.4	5.7 ± 0.3(31)	11%
T2-POET	.28	4	14	--	--	--	--	--	9.4	8.9 ± 0.7(17)	5%
T2-SOS	.23	4	18	--	--	--	--	--	9.5	9.7 ± 0.8(32)	- 3%
T2-DISPED	.24	2	4	2	1	--	--	--	5.6	4.1 ± 0.3(32)	26%
T3-POET	.19	3	12	--	--	--	--	--	6.3	6.3 ± 0.4(24)	0%
T3-SOS	.23	2	7	--	--	--	--	--	4.3	4.0 ± 0.3(37)	8%
T3-DISPED	.23	1	2	1	1	--	--	--	3.3	3.5 ± 0.2(38)	- 7%
T4-POET	.19	13	92	--	--	--	--	--	35.3	37.1 ± 4.3(20)	- 6%
T4-SOS	.23	12	47	--	--	--	--	--	26.8	32.7 ± 1.8(16)	-22%
T4-DISPED	.24	2	6	1	3	--	--	3.8	11.6	14.3 ± 1.1(33)	-23%
T5-MARKUP	.25	--	3.2	--	2.5	4	24.9	--	11.1	10.5 ± 1.1(27)	6%
T5-DRAW	.25	7.6	12.6	--	5	--	--	--	18.9	12.5 ± 3.0(22)	34%
T5-SIL	.27	1	4	0.4	2	--	--	--	4.8	5.4 ± 0.7(32)	-12%
T6-MARKUP	.26	1	7	2	1	--	--	--	5.0	6.2 ± 0.4(34)	-23%
T6-DRAW	.25	1	7	1	1	--	--	--	4.6	5.9 ± 0.4(34)	-29%
T6-SIL	.27	--	6	1.4	1	--	--	--	3.3	3.6 ± 0.3(19)	- 9%
T7-MARKUP	.24	--	8.6	--	4.8	6	13.6	--	15.1	15.0 ± 2.1(29)	2%
T7-DRAW	.19	5	13	--	8	--	--	--	18.0	18.2 ± 1.9(9)	- 1%
T7-SIL	.28	1	8	--	5	--	--	--	9.1	12.3 ± 2.1(23)	-36%
T8-MARKUP	.26	--	8	--	8	1	4.0	--	12.3	9.3 ± 0.4(22)	24%
T8-DRAW	.21	1	5	--	3	--	--	--	5.7	5.3 ± 0.3(25)	7%
T8-SIL	.27	1	5	0.7	2	--	--	--	5.2	4.1 ± 0.2(33)	20%
T9-MARKUP	.25	2	8	--	6.5	--	--	3.5	15.4	13.0 ± 2.5(26)	15%
T9-DRAW	.22	--	5.7	--	5.7	--	--	--	7.5	10.5 ± 1.0(25)	-40%
T9-SIL	.28	--	5	0.3	3	--	--	--	4.8	6.0 ± 1.0(28)	-24%
T10-LOGIN	.29	2	28	--	--	--	--	15.9	27.4[d]	25.1 ± 0.7(29)	9%
T11-FTP	.30	5	31	--	--	--	--	10.1	26.1	19.7 ± 0.7(29)	24%
T12-CHAT	.31	1	11	--	--	--	--	8.3	13.1	11.5 ± 0.6(36)	12%
T13-DIR	.30	2	20	--	--	--	--	0.5	9.2	6.6 ± 0.3(32)	28%
T14-DELVER	.32	2	20	--	--	--	--	0.4	9.4	7.5 ± 0.4(33)	20%

[a]The calculations are done according to formula (1) using the operator times in Figure 1, except for t_K.
[b]t_K is the average time from the typing tests for the subjects on a given system. Each subject's time is weighted by the correct number of instances for that subject on a given task (see Section 3B1).
[c]SE is the standard error of estimation of the population mean for samples of size N.
[d]The execute time for this task also includes .7 sec for the operation of moving the telephone receiver (see Section 3B1).

Figure 5. Calculated and Observed Execution Times in the Experiment

Figure 6. Predicted vs Observed Execution Times in the Experiment



50
40
30
20
15
10
8
6
5
4
3
2
1
Observed execution time (sec)
1
2
3
4
5
6
8
10
15
20
30
40
50
Predicted execution time (sec)
Text editors
POET
SOS
DISPED
Graphics editors
MARKUP
DRAW
SIL
Executive subsystems
All subsystems

the executive subsystem tasks were overpredicted. Examination of the individual observations does not reveal any small set of outliers or particular users that inflate the prediction error.

Prediction accuracy is related to the duration of the attempted prediction. The results above are for single unit tasks. Since unit tasks are essentially independent, prediction of the time to do a sequence of tasks will tend to be more accurate. This can be seen directly in the present data, since each user ran all the tasks for a given system. For example, consider predicting by the model how long it took to do *all four* editing tasks. The average RMS error is only 5%. The corresponding RMS error for the graphics editors over the five tasks is only 6%.

Ideally, all of the parameters of the model should be determined independently of the experimental situation. This was achieved for all the physical operation times, but not for the mental operation time, t_M. We did not have available an appropriate independent source of data from which to determine t_M†. The accuracy of the model is somewhat inflated by the determination of one of its parameters from the data itself. The substantial variability of t_M indicates that this inflation is probably not too serious, which is to say that small changes in the value of t_M don't make much difference. For example, if a t_M as small as 1.2 sec or as large as 2.0 sec were used in the predictions, the RMS error for the Keystroke Model would only increase from 21% to 23%.

The variability of the observed task times is of interest per se, since user behavior is inherently variable. In our data the average coefficient of variation (CV = SD/Mean) of the individual observations over each task is .31, which is the normal variability for behavior of this duration (Card, Moran, & Newell, in press). In comparing the predictions of the model against any actual behavior, the prediction error will always be confounded with some error from the process of sampling the behavior. The sampling error for each of our observed task times is indicated in the SE column of Figure 5. The average standard error is 9%. That the prediction error of the Keystroke Model is over two times larger than this indicates that most of the prediction error is due to a real failure in the model and not just to unreliable observations. It is a totally different issue, of course, whether a 21% prediction error is satisfactory for practical applications.

3B3. Acquisition Time

Turning from the execution part of the task to the acquisition part, the data shows that it took users 2 sec on the average to acquire the tasks from the manuscript. This number may be refined by breaking the tasks into three types: (a) those tasks that the user already had in memory (the executive subsystem tasks that were done each time in

†This is not an inherent difficulty of mental operators; the t_M estimated from this experiment is now available as an independent estimate.

Task Type	Task numbers	Acquisition Time $M \pm SE^a$ (N) (sec) (sec)
All tasks	T1–T14	2.0 ± 2.0 (885)
Repeated task, recalled from memory	T10, T12, T13, T14	0.5 ± 0.3 (130)
Task acquired by looking at manuscript	T1–T4 (POET, SOS), T5–T9, T11	1.8 ± 1.9 (621)
Task acquired by looking at manuscript; then scan for task on display	T1–T4 (DISPED)	4.0 ± 1.9 (134)

[a]SE is the standard error of estimation of the population mean for samples of size *N*.

Figure 7. Observed Acquisition Times in the Experiment

the same order); (b) those tasks for which the user had to look at the manuscript each time (all the graphics tasks, the POET and SOS tasks, and task T11); and (c) those tasks for which the user had to look at the manuscript, then scan text on the CRT to locate the task. The times for these three types of acquisition are given in Figure 7. Users took only .5 sec when the task was in memory, 1.8 sec to get the task from the manuscript, and 4.0 sec to get the task from the manuscript and scan the CRT. These times for the user to acquire a task are similar to results obtained in previous experiments (Card, Moran, & Newell, 1976).

We can use the acquisition times in Figure 7, along with the predicted execution times in Figure 5, to predict the total task times. The RMS error of these predictions is 21%, which is just as accurate as predicting the execution times alone.

4. Sample Applications of the Keystroke Model

The experiment has provided evidence for the Keystroke Model in a wide range of user-computer interactions. Given the method used, the time required for experts to perform a unit task can be predicted to within about 20% by a linear function of a small set of operators. This result is powerful in permitting prediction without any measurements of the actual situation and in expressing the prediction as a simple algebraic expression. The result is narrow in requiring that the method be completely specified at the level of keystrokes and hand motions and in being limited to error-free expert behavior.

In this section we illustrate how the Keystroke Model can be used, both to exploit its power and to work within its restrictions. The basic application—to predict a time for a specific situation by writing down a method and computing the value—has been sufficiently illustrated in the course of the experiment, where such point predictions were made for 32 different tasks involving 10 highly diverse systems. We go on to show three further uses: (1) calculated benchmarks for systems; (2) parametric analysis, where predictions are expressed as functions of task variables; and (3) sensitivity analysis, where changes in the predictions are examined as a function of changes in task or model parameters.

4A. Calculated Benchmarks

Given the ability to predict tasks, it is possible to calculate the equivalent of a benchmark for a system and hence to compare systems. This has obvious cost advantages over obtaining actual measurements. More important, it permits benchmarking at design time, before the system exists in a form that permits measurement. The analysis for the experiment lets us illustrate this easily.

Consider the three text editors, POET, SOS, and DISPED. Let the benchmark be the four tasks T1 to T4. We can use the Keystroke Model to compute the total time to do the benchmark for each system. The answer comes directly from Figure 5 by summing the calculated $T_{execute}$ for T1–T4 for each editor. This gives 59.8 sec, 50.2 sec, and 26.9 sec as the predicted execution times, respectively, for POET, SOS, and DISPED. Taking the POET time (the slowest) as 100, we get ratios of 100:84:45. Thus, as we might have expected, the two line-oriented editors are relatively close to each other and the display editor is substantially faster (about twice as fast). Since we have also done the experiment, we can compare these calculated benchmarks with the observed benchmarks (by summing the observed $T_{execute}$ from Figure 5). We get 60.1 sec, 56.0 sec, and 27.6 sec, respectively. This gives experimentally determined ratios 100:93:46, which is essentially the same result. The agreement between the calculated and observed benchmark provides confidence only in using the calculated benchmark in place of a measured one. It does not provide evidence for the validity of the particular benchmark (tasks T1–T4) or whether benchmarks are generally a valid way to compare editors.

A similar analysis can be performed for the three graphics systems, using tasks T5–T9 as the benchmark. This yields predicted ratios of 100:93:46 for MARKUP, DRAW, and SIL, respectively, with observed ratios of 100:97:58. MARKUP and DRAW are close enough to raise the question of whether the predicted difference between them is too small to be reliable. The calculated difference between MARKUP and DRAW on the benchmark is 59.0–54.7 = 4.3 sec or 7%. The model has an RMS prediction error of 21% for a single unit task. Since this benchmark is essentially an independent sum of five unit tasks, the RMS error should (theoretically) be 21%/SQRT(5) = 9%†. Thus, the predictions for the two systems are within the RMS error of the model, and so the predicted difference between them can hardly be reliable. The fact that the model correctly predicted that DRAW was slightly faster than MARKUP was lucky—there is no reason to expect the Keystroke Model to make such close calls.

4B. Parametric Analysis

Consider the following task: a user is typing text into an editor and detects a misspelled word n words back from the word he is currently typing. How long will take to correct the misspelled word and resume typing?

In DISPED there are two methods for making the correction, which we wish to compare. Since the methods may behave quite differently depending on how far back the misspelled word is, we need to determine how long each method takes as a function of n. The first method makes use of the Backword command (called by hitting the CTRL key and then W), which erases the last typed-in word:

Method W (Backword):

Setup Backword command	MK[CTRL]
Execute Backword n times	$n((1/c)$MK[W])
Type new word	5.5K[word]
Retype destroyed text	5.5(n - 1)K

$$T_{execute} = (1+n/c)t_M + (1+6.5n)t_K \qquad (2)$$
$$= 1.6 + 2.16n \text{ sec}$$

The execution time is a function not only of n, but also of another parameter called c. When a user has to repeat a single-keystroke command several times, such as the Backword command in the above method, he will tend to break the sequence into small

†Recall in Section 3B2 that the actual RMS error for the graphics systems was 6%.

bursts or *chunks*, separated by pauses, which are represented as **M** operations, according to Rule 2 in Figure 2. Thus, we postulate a chunk size, c, which is the average number of Backword commands in a burst. This is used in the second step in the above method, where we count $1/c$ **M** operations for each call of the Backword command. An exact value for c is unknown, but we use a "reasonable" value, $c = 4$, in our calculations (we will return to this decision later in Section 4C). In the calculations we also assume an average non-secretary typist ($t_K = .28$ sec) and an average word length of 5.5 characters (including punctuation and spaces).

The second method is to get out of type-in mode, use the Replace command to correct the word, and then get back into type-in mode, so that input can be resumed:

Method R (Replace):

Terminate type-in mode	MK[ESC]
Point to target word and select it	H[mouse] P[word] K[YELLOW]
Call Replace command	H[keyboard] MK[R]
Type new word	4.5K[word]
Terminate Replace command	MK[ESC]
Point to last input word and select it	H[mouse] P[word] K[YELLOW]
Re-enter type-in mode	H[keyboard] MK[I]

$$T_{execute} = 4t_M + 10.5t_K + 4t_H + 2t_P$$
$$= 12.1 \text{ sec}$$

The predicted time for each method as a function of n is plotted as the solid lines in Figure 8a. As the figure shows, it is faster to use the Backword method up until a certain crossover point, n_{WR}, after which it becomes faster to use the Replace method. Under the above assumptions, the crossover from the Backword method to the Replace method is found to be at 4.9 words.

Suppose a designer wants to add a feature to DISPED to improve performance on this task. We wish to determine, *before* implementation, whether the proposed feature is likely to be much of an improvement.

The designer proposes two new commands. The first is a Backskip command (CRTL S), which moves the insertion point back one word without erasing any text. The second is a Resume command (CTRL R), which moves the insertion point back to the end of the current type-in (where Backskip was first called). These commands allow:

Method S (Backskip):

Setup Backskip command	MK[CTRL]
Execute Backskip $n-1$ times	$(n-1)((1/c)$MK[S]$)$
Call Backword command	MK[W]
Type new word	4.5K[word]
Call Resume command	M2K[CTRL R]

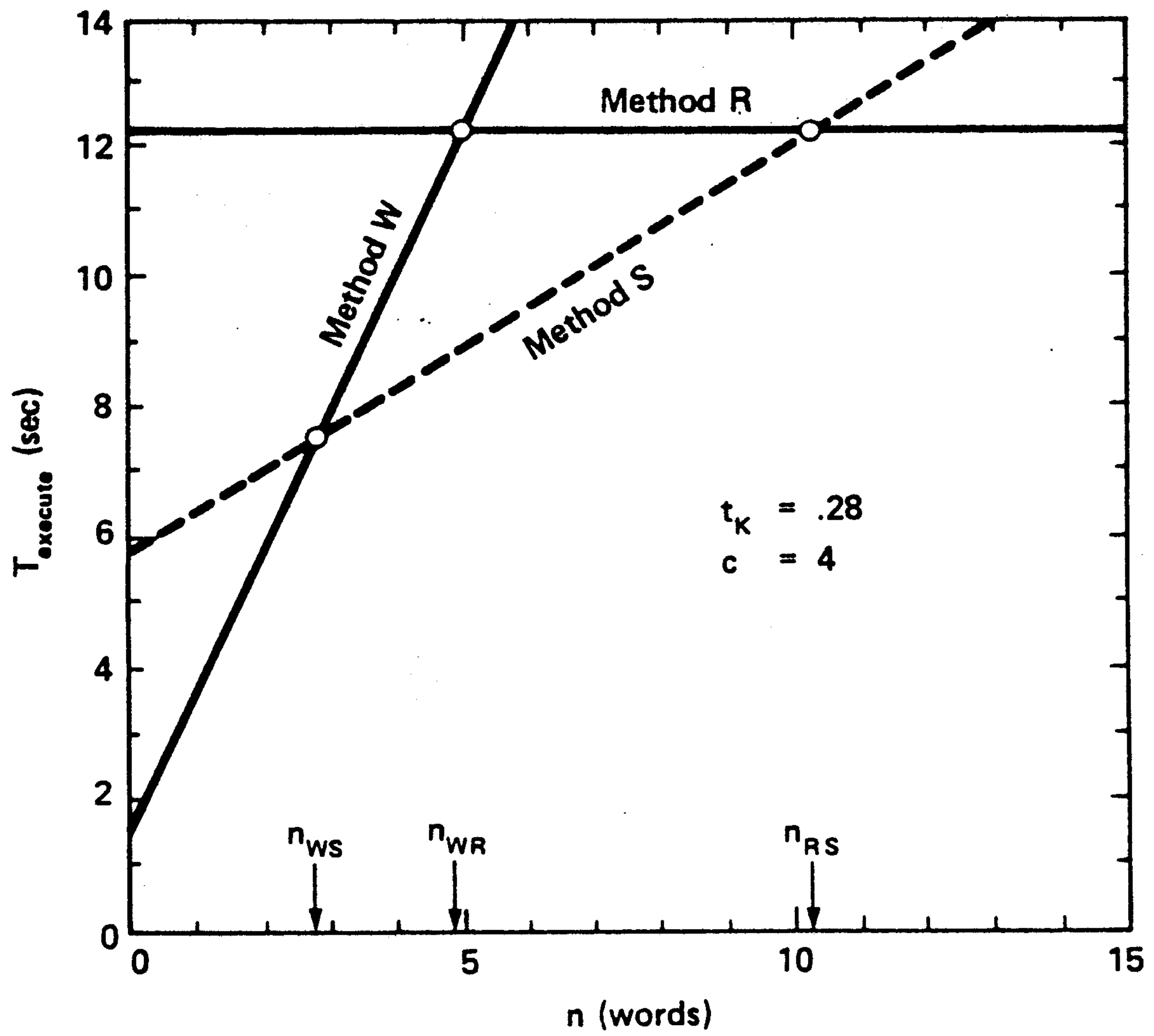
$$T_{execute} = (3+(n-1)/c)t_M + (n+7.5)t_K \quad (3)$$
$$= 5.8 + .62n \text{ sec}$$

The predicted time for the Backskip method is plotted as the dashed line in Figure 8a. With the addition of this method there are two additional crossover points, n_{WS} and n_{RS}, between it and the other two methods. As can be seen, the Backskip method is faster than both of the other methods between n_{WS} and n_{RS}, i.e., from 2.7 to 10.2 words. Thus, a brief analysis provides evidence that the proposed new feature probably will be useful, in the sense that it will be the fastest method over a region of the task space.

4C. Sensitivity Analysis

How sensitive are the calculations above to variations in the parameters of the methods? The question of interest is whether, over such variations, there is still a region in the task space in which the Backskip method is the fastest. An important parameter is the user's typing speed, t_K. How much does the crossover between the Backword method and the Backskip method change as a function of typing speed? Setting equation (2) equal to (3) and solving for n as a function of t_K gives $n = 1.2+.43/t_K$. The crossover increases with faster typists (decreasing t_K), going up to $n = 6.6$ words for the fastest typist ($t_K = .08$ sec). That is to say, faster typists should prefer the old Backword method (which involves more typing) for larger n before switching to the new Backskip method (which involves less typing, but more mental overhead).

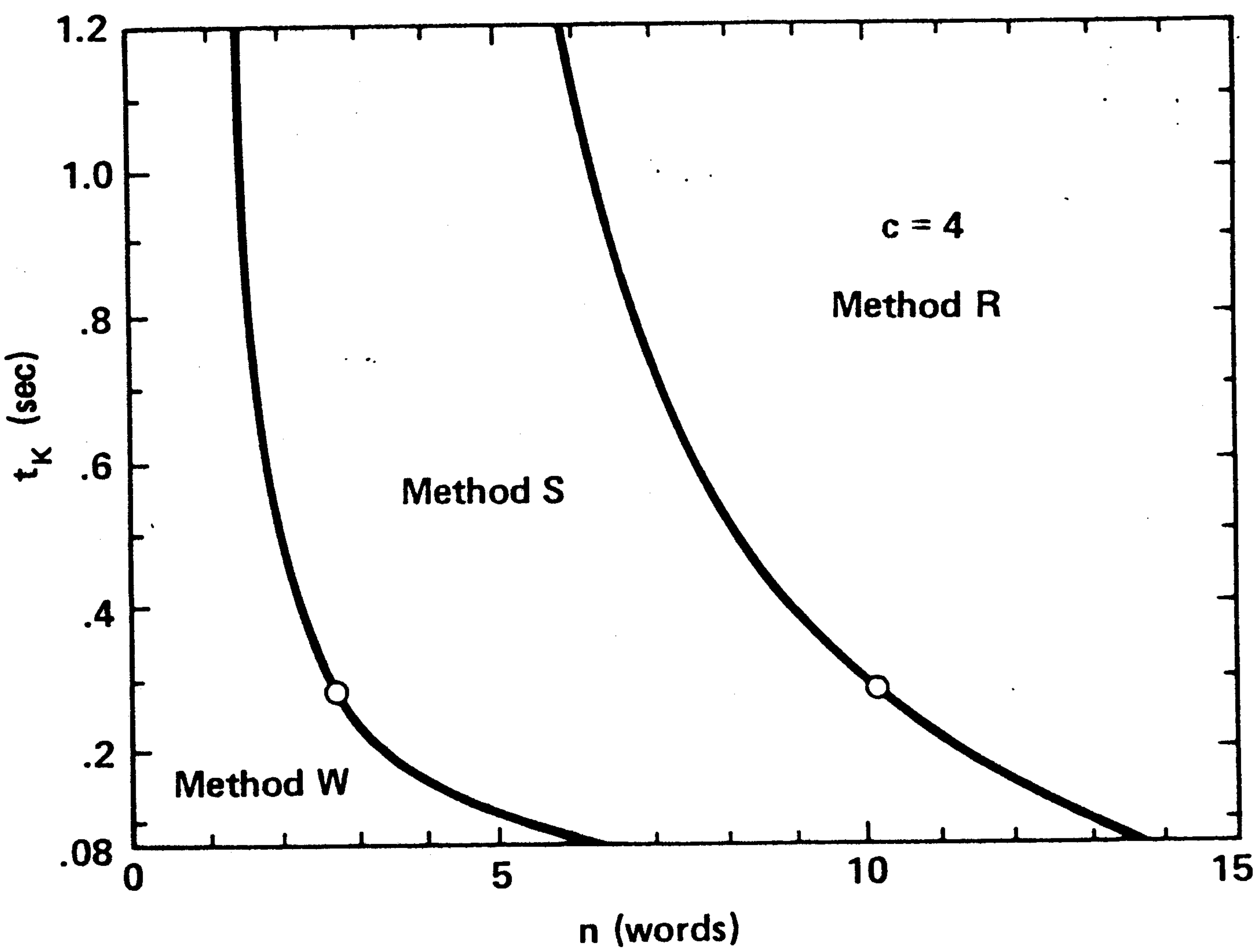
Figure 8a. Execution Time for Three Methods as a Function of *n*

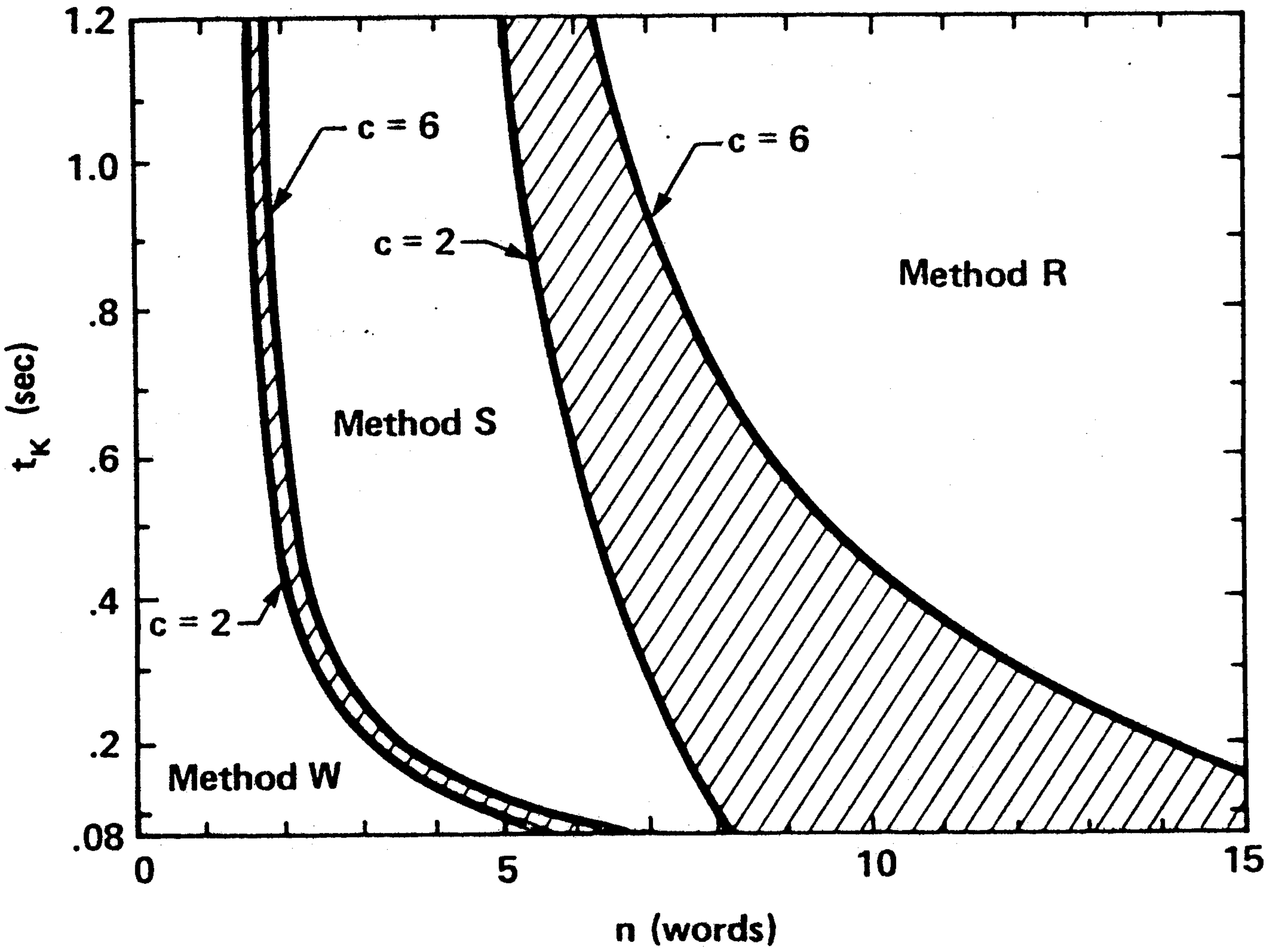


Method R
Method W
Method S
t_K = .28
c = 4
$T_{execute}$ (sec)
n (words)
n_{WS}
n_{WR}
n_{RS}
0
2
4
6
8
10
12
14
0
5
10
15

Figure 8b. Phase Diagram for the Fastest Method

Figure 8c. Phase Diagram Adjusted for Different Chunk Sizes


1.2
1.0
.8
.6
.4
.2
.08
t_K (sec)
0
5
10
15
n (words)
c = 4
Method R
Method S
Method W



1.2
1.0
.8
.6
.4
.2
.08
t_K (sec)
0
5
10
15
n (words)
c = 6
c = 2
c = 2
c = 6
Method W
Method S
Method R

We can plot the crossover boundary between the two methods in the space of the two parameters: n (characterizing different tasks) and t_K (characterizing different users). The two boundaries of the new Backskip method are plotted in Figure 8b. These boundaries define the regions in the parameter space where each method is fastest, that is, a phase diagram for the fastest method. The circles mark the crossover points corresponding to the ones in Figure 8a (i.e., at $t_K = .28$ sec). This diagram clearly shows the shift of crossovers for fast typists. It also shows that, for any speed of typist, there are some tasks for which the Backskip method is the fastest.

We are not sure of the exact chunk size, c. Thus, we must check whether our conclusions about the usefulness of the Backskip method are sensitive to the choice of a value for c. To do this, we re-derive the crossover between the Backword and Backskip methods by setting equation (2) equal to (3) and solving for n as a function of both c and t_K; this gives $n = 1.2+.49/t_K-.24/ct_K$. Although we don't know an exact value for c, we can be reasonably confident that it will stay between 2 and 6. With $t_K = .28$ sec, for example, the crossover varies between 2.5 and 2.8 words as c varies between 2 and 6; so the value of c doesn't seem to have a great effect at this point.

The best way to see the overall affect of the value of c is to re-plot the phase diagram using the reasonable extreme values of c. The two crossover boundaries for the Backskip method are plotted in Figure 8c as "fat" lines defined by setting c to 2 and to 6 in the crossover equations. This diagram clearly shows that the value of c affects one boundary more than the other. The boundary between the Backword and Backskip methods is not affected much by c, because the chunk size is involved in both methods in exactly the same way. But the boundary between the Backskip and Replace methods is greatly affected by the value of c, since c is not involved in the Replace method at all. Small chunk sizes, especially, penalize the Backskip method. Overall, however, varying c does not squeeze out the region for the Backskip method; and our basic conclusion—that the new method is a useful addition—still holds.

There are other aspects of the above methods for which we could do a sensistivity analysis. (For example, if the last two **M** operations of the Backskip method were eliminated according to Rule 1, how much would the value of the Backskip method increase?) However, the sensitivity analyses above illustrate how the Keystroke Model can be used to evaluate design choices—even when many aspects of the calculation are uncertain—for the principal conclusions are often insensitive to many of the uncertainties.

5. Simplifications of the Keystroke Model

The question naturally arises as to whether further simplifications of the Keystroke Model might do reasonably well at predicting execution time. One could (a) count only the number of keystrokes or (b) count all the physical operators and prorate the time for mental activity or (c) use a single constant time for all operators. We show below that such simplifications substantially degrade accuracy. However, they provide useful approximations where the lowered accuracy can be tolerated.

5A. Keystrokes Only

With this simplification, execution time is proportional to the number of keystrokes:†

$$T_{execute} = \kappa n_K + T_R .$$

We separate out the system response times (T_R) so as not to confound the comparison. The constant of proportionality κ is to be distinguished from t_K, the typing speed, which is determined from standard typing tests. Estimating the value of κ from a least squares fit of the values of n_K and the observed $T_{execute}$ in Figure 5 gives $\kappa = .49$ sec/keystroke. The correlation between the times predicted by this model and the observed times is .87, and the RMS error is 49%. The statistics for comparing all models are presented in Figure 9. As can be seen, using keystrokes only is substantially less accurate than the full Keystroke Model. This simplification is inappropriate for tasks that are not dominated by keystroking. For example, it only predicts about a third of the observed time for the MARKUP tasks, which are dominated by pointing and drawing operations.

The above estimate of κ is held down by one outlying point in the data, T4-POET ($n_K = 92$). Estimating κ with this one point removed gives $\kappa = .60$ sec, a value close to another estimate obtained in an earlier benchmark study (Card, Moran, & Newell, in press). T4-POET is the only task that requires any input-typing of text. One obvious refinement of the keystrokes-only model would be to distinguish two kinds of keystrokes: mass input-typing (at t_K sec/keystroke) versus command-language keying (at κ sec/keystroke). For this purpose, a κ of .60 sec is the more reasonable value.

†The keystrokes-only model is similar to the one proposed by Embly et al. (1978). We cannot compare our results to theirs, however, since they did not test their model against any empirical data.

Model Variation	Parameters	Correlation (r)[a]	RMS Error[b]
Keystrokes Only	κ = .49 sec/keystroke[c]	.87	49%
Prorated Mental Time	μ = 1.67	.81	45%
Constant Operator Time	τ = .43 sec/operator[c]	.92	34%
Keystroke Model	(See Figure 1)	.95	22%

[a]The correlations are between the execution times predicted by each of the models and the observed execution times from Figure 5.
[b]The RMS error is given as a percentage of the observed execution time, which was 11.0 sec.
[c]More useful parameter values are κ = .60 sec and τ = .49 sec (see Sections 5A and 5C).

Figure 9. Comparison of the Keystroke Model With Simpler Variations

5B. Prorated Mental Time

According to this simplification, execution time is the time required for the physical operations multiplied by a factor to account for the mental time:

$$T_{execute} = \mu\,(T_K + T_H + T_P + T_D) + T_R\,.$$

The idea behind this simplification is that on the average the various physical operations will require a constant overhead of mental activity. Thus, instead of trying to predict exactly how many mental operations there are (as we do in the full Keystroke Model), we can do fairly well by just using a multiplicative mental overhead constant, μ.

Using a least squares analysis to determine μ from the sum of the calculated times for the physical operations and the observed values of $T_{execute}$ in Figure 5 gives μ = 1.67; i.e., there is a 67% overhead for mental activity. The correlation between predicted and observed times is .81 and the RMS error is 45%.

This simplification is also less accurate than the Keystroke Model, as can be seen in Figure 9. This suggests that the extra detail in the Keystroke Model, involving the placements of the mental preparedness operator, **M**, is effective. It is this operator that qualifies the Keystroke Model as a genuine psychological model and not simply as an analysis of the physical operations, as in a standard time and motion study.

There is an interesting relation between these simpler models and the rules for placing occurrences of **M** in the Keystroke Model (Figure 2). The initial placement (by Rule 0) of **M**'s with certain **K**'s and **P**'s is essentially an assumption that mental time is proportional to a subset of the physical operators. If Rule 0 had specified all physical operators, Rule 0 by itself would have been equivalent to prorating mental time. If the other physical operators (**P**, **H**, and **D**) had been ignored, this would have been equivalent to counting keystrokes only. Thus, the deletion of the **M**'s according to rules Rules 1 to 4 constitute the ways in which the Keystroke Model departs from these simpler models. The evidence for the superiority of the Keystroke Model presented in Figure 9 is also evidence that rules Rules 1 to 4 had a significant effect. In fact, each of the rules individually makes a significant contribution, in the sense that its removal leads to a decrease in the accuracy of the Keystroke Model.

5C. Constant Operator Time

According to this simplification, execution time is proportional to the *number* of Keystroke Model operations:

$$T_{execute} = \tau\,(n_M + n_K + n_P + n_H + n_D) + T_R\,.$$

The idea behind this simplification is the statistical observation (Wainer, 1976) that the accuracy of linear models is not sensitive to the differential weighting of the factors—equal weighting does nearly as well as any other weighting. Thus, we disregard the different operator times and use a single time, τ, for all operators. Note that the constant-operator-time model is formally similar to the keystrokes-only model; the latter can be viewed as using n_K as a crude estimate of the total number of operators.

Estimating τ by a least squares fit of the data in Figure 5 gives τ = .43 sec/operator. The correlation between predicted and observed times is .92, and the RMS error is 34%. (For the same reason discussed in Section 5A, it is useful to estimate τ with the T4-POET task removed, getting τ = .49 sec/operator.)

The constant-time model is quite a bit more accurate than the keystrokes-only model, which tells us that taking into account operators other than **K** is useful. In fact, most of the action in the constant-time model (over the set of data in Figure 5, at least) comes from counting only the **K**, **P**, and **M** operators. In any particular task, of course, any of the operators can be dominant. On the other hand, the constant-time model is still less accurate than the Keystroke Model, showing that taking into account accurate estimates of each operator's time yields another increment of accuracy.

In summary, all of the simplifications presented in this section are less accurate than the Keystroke Model. However, these simplified models are probably good enough for many practical applications, especially for "back-of-the-envelope" style calculations, where it is too much trouble to worry about the subtleties of counting the **M**'s that the Keystroke Model requires.

6. Conclusion

We have presented the Keystroke Model for predicting one important aspect of the total user-computer interaction, namely, the time to perform a task. The Keystroke Model has several requirements: the user must be an expert; the task must be a routine unit task; the method must be specified in detail; and the performance must be error-free. Yet, the model appears to be sufficiently powerful and flexible to be widely applicable.

The Keystroke Model is a *system design tool.* We have had two main concerns in shaping this tool. First, the tool must be quick and easy to use, if it is to be useful *during* the design of interactive systems. The existing strengths of psychology and human factors methods are primarily in the design and analysis of experiments; but experiments are too slow and cumbersome to be incorporated in practice. This concern implies that the tool be analytical—one that permits calculation in the style familiar to all engineers. Second, the tool must be useful to practicing computer system designers, who are not psychologists. This implies that the entire tool must be packaged to avoid requiring specialized psychological knowledge. We think that the Keystroke Model satisfies these concerns, along with the primary consideration of being accurate enough to be worth using and worth playing with to find new ways to use it. We believe the Keystroke Model belongs in the tool-kit of all practicing system designers.

More complicated and refined models than the Keystroke Model can be formulated, which relax some of its serious restrictions, such as models that predict the method used or models that apply to non-experts. One of the great virtues of the Keystroke Model, from our own perspective as scientists trying to understand how humans interact with computer systems, is that it puts a lower bound on the effectiveness of new proposals. Any new proposal must now do better than the Keystroke Model to command serious consideration.

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Appendix

Methods for All the Experimental Tasks

This Appendix lays out the methods for all the task-system combinations used in the experiment described in Section 3.

The conventions for encoding methods in terms of the Keystroke Model operations is explained in Section 2. The following notes explain specific points in the method encodings. Citations to these notes appear in the column at the extreme right.

1.	.3P	Fractional coefficients indicate that certain actions occurred less than 100% of the time. The order of tasks influences the necessity of certain actions, such as pen or command selection. In addition, the way in which pen selection is done (in MARKUP) influences the frequency of certain actions.
2.	0H	The coefficient of zero indicates this action does not occur in this task.
3.	R(*t*)	A response time of specified seconds indicates that the response time has been measured and added to the task time.
4.	R(0)	A response time of zero indicates that the actual response time is absorbed in the beginning of the subsequent task and therefore not added to the task time of the current task.
5.	7.0K	The search string averaged 7 characters, therefore 7.0K represents the average—not absolute—number of keystrokes involved.
6.	52.0K	This number indicates an average sentence length of 52 characters.
7.	D(*n*,*l*)	*n* is the number of segments drawn, and *l* is the total length of all segments drawn (in centimeters).
6.	M18C	This is a special operator taken from the MTM predetermined time standards (Maynard, 1971).

Task T1: Replace one 5-letter word with another (one line from previous task)

T1-POET:	
SELECT WORD	
Linefeed to location	M K[LF]
REPLACE WORD	
Issue Substitute command	M K[S]
Type new word	5K[word]
Terminate new word	M K[RETURN]
Type old word	5K[word]
Terminate old word	M K[RETURN]
Terminate command	K[RETURN]
T1-SOS:	
REPLACE WORD	
Issue Substitute command	M K[S]
Type old word	5K[word]
Terminate old word	M K[ESC]
Type new word	5K[word]
Terminate new word	M K[ESC]
Type line number	5K[number]
Terminate command	M K[RETURN]
T1-DISPED:	
SELECT WORD	
Reach for mouse	H[mouse]
Point to word	P[word]
Select word	K[YELLOW]
REPLACE WORD	
Home on keyboard	H[keyboard]
Issue Replace command	M K[R]
Type new word	5K[word]
Terminate command	M K[ESC]
Wait for completion	R(0) 4

Task T2: Add a fifth character to four character word (one line from previous task)

T2-POET:	
SELECT WORD	
Linefeed to location	M K[LF]
Issue Substitute command	M K[S]
REPLACE WORD	
Type new word	5K[word]
Terminate new word	M K[RETURN]
Type old word	4K[word]
Terminate old word	M K[RETURN]
Terminate command	K[RETURN]

T2-SOS:

REPLACE WORD		
Issue Substitute command	M K[S]	
Type old word	4K[word]	
Terminate old word	M K[ESC]	
Type new word	5K[word]	
Terminate new word	M K[ESC]	
Type line number	5K[word]	
Terminate command	M K[RETURN]	

T2-DISPED:

SELECT WORD		
Reach for mouse	H[mouse]	
Point to word	P[word]	
Select word	K[YELLOW]	
APPEND CHARACTER		
Issue Append command	M K[A]	
Home on keyboard	H[keyboard]	
Type additional letter	K[character]	
Terminate command	M K[ESC]	
Wait for completion	R(0)	4

Task T3: Delete a line, all on one line (eight lines from previous task)

T3-POET:

FIND LINE		
Indicate search string	M K[QUOTE]	
Type search string	7.0K[string]	5
Terminate search string	M K[QUOTE]	
Print line	K[SLASH]	
DELETE LINE		
Issue Delete command	M K[D]	
Terminate command	K[RETURN]	

T3-SOS:

DELETE LINE		
Issue Delete command	M K[D]	
Type 5-digit line number	5K[number]	
Terminate command	M K[RETURN]	

T3-DISPED:

SELECT LINE		
Reach for mouse	H[mouse]	
Point to line	P[line]	
Select line	K[RED]	
DELETE LINE		
Issue Delete command	M K[D]	
Wait for completion	R(0)	4

Task T4: Move a 50-character sentence (on two lines) to end of paragraph (eight lines from previous task)

T4-POET:		
FIND TEXT		
Indicate search string	M K[QUOTE]	
Type search string	7.0K[string]	5
Terminate search string	M K[QUOTE]	
Print line	K[SLASH]	
EDIT LINE 1		
Issue Edit command	M 2K[E RETURN]	
Issue Search subcommand	M 2K[CTRL S]	
Type unique capital letter	2K[SHIFT character]	
Terminate command	M K[RETURN]	
EDIT LINE 2		
Linefeed to next line	M K[LF]	
Issue Edit command	M 2K[E RETURN]	
Issue subcommand to delete first part of line	M 2K[CTRL Y]	
Type unique capital letter	2K[SHIFT character]	
Terminate command	M 2K[CTRL Z]	
FIND TEXT		
Indicate search string	M K[QUOTE]	
Type search string	7.0K[string]	5
Terminate search string	M K[QUOTE]	
Print line	K[SLASH]	
MOVE TEXT		
Issue Append command	M 2K[A RETURN]	
Type sentence	52.0K[sentence]	6
Terminate type-in	M 2K[CTRL Z]	
T4-SOS:		
EDIT LINE 1		
Issue Alter command	M K[A]	
Type 5-digit line number	5K[number]	
Confirm	M K[RETURN]	
Issue Search subcommand	M K[S]	
Type unique capital letter	2K[SHIFT character]	
Issue Insert subcommand	M K[I]	
Type new text	K[RETURN]	
Terminate subcommand	M K[ESC]	
Terminate command	K[RETURN]	
EDIT LINE 2		
Issue Alter command	M K[A]	
Type 5-digit line number	5K[number]	
Confirm	M K[RETURN]	
Issue Search subcommand	M K[S]	
Type unique capital letter	2K[SHIFT character]	
Issue Insert subcommand	M K[I]	
Type new text	K[RETURN]	
Terminate subcommand	M K[ESC]	
Terminate command	K[RETURN]	

MOVE TEXT		
Issue Transfer command	M K[T]	
Type 5-digit line number for new location	5K[number]	
Type separator	K[COMMA]	
Type first 5-digit line number	5K[number]	
Type separator	K[COLON]	
Type last 5-digit line number	5K[number]	
Terminate command	M K[RETURN]	

T4-DISPED:

DELETE TEXT		
Reach for mouse	H[mouse]	
Point to beginning of deletion	P[text]	
Select beginning point	K[RED]	
Point to end of deletion	P[text]	
Select ending point	K[BLUE]	
Issue Delete command	M K[D]	
Wait for deletion	R(3.8)	3
MOVE TEXT		
Point to new location	P[text]	
Select new location	K[RED]	
Issue Append command with default	M 2K[A ESC]	
Wait for completion	R(0)	4

Task T5: Add a box to a diagram

T5-MARKUP:

SELECT PEN		
Reach for mouse	0H[mouse]	2
Point to clear place	.6P[display]	1
Press and hold YELLOW mouse button	.6K[YELLOW]	1
Point to expand menu	.3P[menu]	1
Point to menu icon	.6P[icon]	1
Release YELLOW mouse button	.6K[YELLOW]	1
DRAW SIDES		
Point to location where corner is to be	P[corner]	
Begin drawing	K[RED]	
Draw 4 sides	D(4,24.86)	7
Terminate drawing	K[RED]	

T5-DRAW:

SELECT COMMAND		
Begin Draw mode	.6M .6K[ESC]	1
DRAW SIDE 1		
Reach for mouse	0H[mouse]	2
Point to location where corner is to be	P[corner]	
Select beginning point	K[BLUE]	
Point to end point	P[corner]	
Select end point	K[BLUE]	
Terminate line definition	M K[ESC]	

DRAW SIDE 2		
Select beginning point	M K[BLUE]	
Point to end point	P[corner]	
Select end point	K[BLUE]	
Terminate line definition	M K[ESC]	
DRAW SIDE 3		
Select beginning point	M K[BLUE]	
Point to end point	P[corner]	
Select end point	K[BLUE]	
Terminate line definition	M K[ESC]	
DRAW SIDE 4		
Select beginning point	M K[BLUE]	
Point to end point	P[corner]	
Select end point	K[BLUE]	
Terminate line definition	M K[ESC]	

T5-SIL:

DRAW BOX		
Reach for mouse	.4H[mouse]	1
Point to location where corner is to be	P[corner]	
Select	K[RED]	
Point to location for opposite corner	P[corner]	
Select	K[BLUE]	
Terminate box definition	M 2K[CTRL B]	

Task T6: Add a 5-character label to a box

T6-MARKUP:

TYPE LABEL		
Home on keyboard	H[keyboard]	
Hit lockshift for capitalized label	M K[LOCK]	
Type 4 character label and release lockshift	5K[word LOCK]	
PUT LABEL		
Reach for mouse	H[mouse]	
Point to location of label	P[location]	
Paste label	K[RED]	

T6-DRAW:

TYPE LABEL		
Home on keyboard	H[keyboard]	
Type word (4 letters, lst capitalized)	5K[word]	
Terminate input string	M K[RETURN]	
PUT LABEL		
Point to location of label	P[display]	
Paste label	K[RED]	

T6-SIL

SELECT LOCATION		
Reach for mouse	.4H[mouse]	1
Point to location of label	P[display]	
Select	K[RED]	
TYPE LABEL		
Home on keyboard	H[keyboard]	
Type label (4 letters, lst capitalized)	5K[word]	

Task T7: Reconnect a 2-stroke line to a different box

T7-MARKUP:

SELECT PEN		
Reach for mouse	0H[mouse]	2
Point to clear place	.3P[display]	1
Press and hold YELLOW mouse button	.3K[YELLOW]	1
Point to expand menu	.15P[menu]	1
Point to menu icon	.3P[icon]	1
Release YELLOW mouse button	.3K[YELLOW]	1
DELETE OLD SEGMENTS		
Point to beginning of segment	P[segment]	
Press BLUE mouse button	K[BLUE]	
Trace line segments	D(2,5.65)	7
Release BLUE mouse button	K[BLUE]	
FIXUP BOX 1		
Point to damaged segment in BOX 1	P[segment]	
Press RED mouse button	K[RED]	
Redraw damaged segment	D(1,1.13)	7
Release RED mouse button	K[RED]	
FIXUP BOX 2		
Point to damaged segment in BOX 2	P[segment]	
Press RED mouse button	K[RED]	
Redraw damaged segment	D(1,1.13)	7
Release RED mouse button	K[RED]	
DRAW NEW SEGMENTS		
Point to where segment is to start	P[segment]	
Press RED mouse button	K[RED]	
Draw	D(2,5.65)	7
Release RED mouse button	K[RED]	

T7-DRAW:

SELECT COMMAND		
Reach for mouse	0H[mouse]	2
Point to delete command	P[icon]	
Select	K[RED]	
DELETE SEGMENT 1		
Point to segment 1	P[segment]	
Delete	K[RED]	
DELETE SEGMENT 2		
Point to segment 2	P[segment]	
Delete	K[RED]	
SELECT COMMAND		
Start DRAW command	M K[ESC]	
FIXUP SEGMENT		
Point to start of segment	P[segment]	
Select	K[BLUE]	
Point to end of segment	P[segment]	
Select	K[BLUE]	
Terminate segment definition	M K[ESC]	

DRAW NEW SEGMENT 1		
Point to start of new segment 1	P[segment]	
Select	K[BLUE]	
Point to end of new segment 1	P[segment]	
Select	K[BLUE]	
Terminate segment definition	M K[ESC]	
DRAW NEW SEGMENT 2		
Select (beginning same as end of last)	M K[BLUE]	
Point to end of new segment 2	P[segment]	
Select	K[BLUE]	
Terminate segment definition	M K[ESC]	

T7-SIL:

SELECT SEGMENT		
Reach for mouse	0H[mouse]	
Point to segment	P[segment]	
Select	K[BLUE]	
MOVE SEGMENT		
Point to new place	P[location]	
Shrink one segment by moving it	2K[CTRL RED]	
DELETE SEGMENT		
Delete other segment	M 2K[CTRL D]	
DRAW NEW SEGMENTS		
Point to start of new segment	P[segment]	
Select	K[RED]	
Point to end of segment	P[segment]	
Draw segment	K[YELLOW]	
Point to end of second segment	P[segment]	
Draw segment	K[YELLOW]	

Task T8: Delete a box, but keep an overlapped line

T8-MARKUP:

SELECT PEN		
Reach for mouse	0H[mouse]	2
Point to clear place	P[display]	
Press and hold YELLOW mouse button	K[YELLOW]	
Point to expand menu	.5P[menu]	1
Point to menu icon	P[icon]	
Release YELLOW mouse button	K[YELLOW]	
DELETE BOX		
Point outside corner of figure	P[corner]	
Press and hold BLUE mouse button	K[BLUE]	
Move mouse past opposite corner	P[corner]	
Release BLUE mouse button	K[BLUE]	
SELECT PEN		
Point to clear place	P[display]	
Press and hold YELLOW mouse button	K[YELLOW]	
Point to expand menu	.5P[menu]	1
Point to menu icon	P[icon]	
Release YELLOW mouse button	K[YELLOW]	

FIXUP SEGMENT		
Point to damaged segment	P[segment]	
Press and hold RED mouse button	K[RED]	
Trace segment	D(1,3.955)	7
Release RED mouse button	K[RED]	

T8-DRAW:

SELECT COMMAND		
Reach for mouse	0H[mouse]	2
Point to 'area select' command	P[icon]	
Select	K[RED]	
SELECT BOX		
Point to corner of figure to be deleted	P[corner]	
Press and hold mouse button	K[RED]	
"Swish" diagonally thru figure to be deleted	P[corner]	
Release mouse button	K[RED]	
DELETE BOX		
Delete from keyboard with control d	M 2K[CTRL D]	

T8-SIL:

SELECT BOX		
Reach for mouse	.7H[mouse]	1
Point just outside corner of figure	P[corner]	
Press RED mouse button	K[RED]	
Point just outside opposite corner	P[corner]	
Select	2K[CTRL BLUE]	
DELETE BOX		
Issue delete command	M 2K[CTRL D]	

Task T9: Copy a box

T9-MARKUP:

SELECT PEN		
Reach for mouse	0H[mouse]	2
Point to clear place	P[display]	
Press and hold YELLOW mouse button	K[YELLOW]	
Point to expand menu	.5P[menu]	1
Point to menu icon	P[icon]	
Release YELLOW mouse button	K[YELLOW]	
DELETE BOX		
Point outside corner of figure	P[corner]	
Press and hold BLUE mouse button	K[BLUE]	
Move mouse past opposite corner	P[corner]	
Release BLUE mouse button	K[BLUE]	
Wait for deletion	R(1.1)	3
RESTORE BOX		
Press and hold RED mouse button	M K[RED]	
Move mouse until figure aligned	P[display]	
Release RED mouse button	K[RED]	
Wait for figure to be pasted	R(1.2)	3

PASTE COPY		
Press and hold RED mouse button	M K[RED]	
Move mouse until figure aligned	P[display]	
Release RED mouse button	K[RED]	
Wait for figure to be pasted	R(1.2)	3
T9-DRAW:		
SELECT COMMAND		
Reach for mouse	0H[mouse]	2
Point to 'area select' command	.7P[icon]	1
Select	.7K[RED]	1
SELECT BOX		
Point to corner of figure to be moved	P[corner]	
Press and hold mouse button	K[RED]	
"Swish" diagonally thru area to be copied	P[corner]	
Release mouse button	K[RED]	
SELECT COMMAND		
Point to 'copy' command	P[icon]	
Select	K[RED]	
COPY BOX		
Point to reference point in old location	P[display]	
Select	K[BLUE]	
Point to reference point in new location	P[display]	
Select	K[BLUE]	
T9-SIL:		
SELECT BOX		
Reach for mouse	.3H[mouse]	1
Point just outside corner of figure	P[corner]	
Select	K[RED]	
Point just outside opposite corner	P[corner]	
Select	2K[CTRL BLUE]	
COPY BOX		
Point to new location	P[display]	
Copy figure	2K[CTRL YELLOW]	

Task T10: Phone computer and log in (4 character name, 6 character password)

DIALUP		
Push 8 digits	8K	
Listen/wait for computer signal	R(1.8)	
LOG IN		
Put phone on terminal cradle	M18C(.7)	8
Watch/wait for carrier signal light	R(.9)	
Type login prompt	M 2K[CTRL C]	
Watch/wait for system greeting	R(5.9)	
Type login (log sp 4K sp 6K sp [1])	17K	
Terminate login	M K[RETURN]	
Watch/wait for login acknowledgement	R(7.3)	

Task T11: Transfer a file to another computer, renaming it

CONNECT COMPUTER		
Issue connect command	4K[F T P SPACE]	
Specify 5-character file-server	5K	
Confirm	M K[RETURN]	
Wait for connection	R(4.1)	3
TRANSFER FILE		
Issue Store command	M 3K[S T SPACE]	
Type old 4-character filename	4K	
Terminate filename	M K[SPACE]	
Type new 10-character filename	10K	
Terminate command	M K[RETURN]	
Wait for completion	R(1.4)	3
CLOSE CONNECTIONS		
Issue Quit command	M K[Q]	
Confirm	K[RETURN]	
Wait for reconnect to local computer	R(4.6)	3

Task T12: Connect to another computer

CONNECT COMPUTER		
Issue Chat command	5K[C H A T SPACE]	
Specify 5-character computer name	5K	
Confirm	M K[RETURN]	
Wait for connection	R(8.3)	3

Task T13: Display a subset of the file directory with file lengths

DISPLAY DIRECTORY		
Issue Directory command	4K[D I R SPACE]	
Specify 10-character filename	10K	
Enter subcommand mode	M K[COMMA]	
Confirm	K[RETURN]	
Issue Length command	2K[LE]	
Confirm	M K[RETURN]	
Terminate command	K[RETURN]	
Wait for completion	R(.5)	3

Task T14: Delete old versions of a file

DELETE OLD VERSION		
Issue Delver command	6K[D E L V E R]	
Confirm	M K[RETURN]	
Answer first system question	K[Y]	
Answer second system question	K[Y]	
Type 10-character filename	10K	
Terminate command	M K[RETURN]	
Wait for completion	R(.4)	3